PRODUCT USER MANUAL

For Antarctic Time series of ice drift

SEAICE_ANT_PHY_L3_MY_011_018

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TABLE OF CONTENTS

GLOSSARY AND ABBREVIATIONS 4
DATA ACCESS 5
I INTRODUCTION 7
II PRODUCT DESCRIPTION 8
II.1 General information 8
II.2 Details of datasets 9
III FILE FORMAT 11
IV ADDITIONAL INFORMATION ABOUT PRODUCTS 12
IV.1 Nomenclature of products 12
IV.2 Grid 12
IV.3 Domain coverage 13
IV.4 Update Time 13
IV.5 Temporal extend of analysis and forecast stored on delivery mechanism 14
IV.6 Other information: mean centre of product, land mask value, missing value 14
V VALIDATION AND ASSESSMENT/ QUALITY CONTROL DONE ON PRODUCTS 15
Structure of files 15

GLOSSARY AND ABBREVIATIONS

TAC	Thematic Assembly Centre
SIW	Sea Ice and Wind
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
RMS	Root mean square
SDN	SeaDataNet (climatology)
PC	Production Centre
PU	Production Unit
CMEMS	Copernicus Marine Environment Monitoring Service

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

This guide describes the regional sea ice product SEAICE_ANT_PHY_L3_MY_011_018 from the Copernicus Marine Ocean and Sea Ice Thematic Assembly Centre (OSI TAC), what data services are available to access them, and how to use the files and services.
These products are produced and disseminated by IFREMER. It is a time series of sea ice drift observation from satellites in the central Antarctic area at medium resolution.

II PRODUCT DESCRIPTION

II.1 General information

Product Lines	SEAICE_ANT_PHY_L3_MY_011_018
Geographical coverage	-180°W 180°E ; -39°N -89.9 °N
Variables	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity
Analysis	No
Forecast	No
Available time series	From April 2003 – present (AMSR) From June 2020 – October 2022 (CFOSAT)
Temporal resolution	24hr average field
Target delivery time	Monthly (automatic update) at month - 2
Delivery mechanism	MDS
Horizontal resolution	0.25 degree (AMSR) 0.5 degree (CFOSAT)
Number of vertical levels	1
Format	Netcdf CF1.4

Table 1: SEAICE_ANT_PHY_L3_MY_011_018 production line

II.2 Details of datasets

SEAICE_ANT_PHY_L3_MY_011_018		
DATASETS	VARIABLES AND UNIT or GEOGRAPHICAL COVERAGE	NAME OF VARIABLES IN THE NETCDF FILE or DOMAIN AREA
cmems_obs-si_ant_physic_my_drift-amsr_P2D	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity
cmems_obs-si_ant_physic_my_drift-amsr_P3D	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity
cmems_obs-si_ant_phy_my_drift-cfosat_P2D	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity

cmems_obs-si_ant_phy_my_drift-cfosat_P3D	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity
cmems_obs-si_ant_phy_my_drift-cfosat_ssmi_merged_P3D	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity
cmems_obs-si_ant_phy_my_drift-cfosat_ssmi_merged_P30D	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity

III FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV ADDITIONAL INFORMATION ABOUT PRODUCTS

IV.1 Nomenclature of products

The nomenclature used for the SEAICE_ANT_PHY_L3_MY_011_018 products is:

<date1>-<date2>_<freqflag><avgflag>-<producer>-<level>-<parameter>-<config>-<region>-<creationdate>-<version>.nc

with

date1= start date

date2=end date

DATE:YYYMMDD

freqflag = d for daily (2d, 3d) (AMSR) / (2d, 3d or 30d) (CFOSAT)

avgflag= m for mean

producer=producer

level = product level (4 for Merging)

parameter= ICEM

config=sensor_drift

region=ANT for Antarctic

creationdate= creation date of the file

version=product version

IV.2 Grid

SEAICE_ANT_PHY_L3_MY_011_018 product is made to have a regular latitude and longitude grids. The dataset parameters are centered in the box of this grid.

IV.3 Domain coverage

The spatial coverage of the SEAICE_ANT_PHY_L3_MY_011_018 product is suitable for the Antarctic sea ice area observations. Here an example of the coverage. Grids type is the stereographic projection.

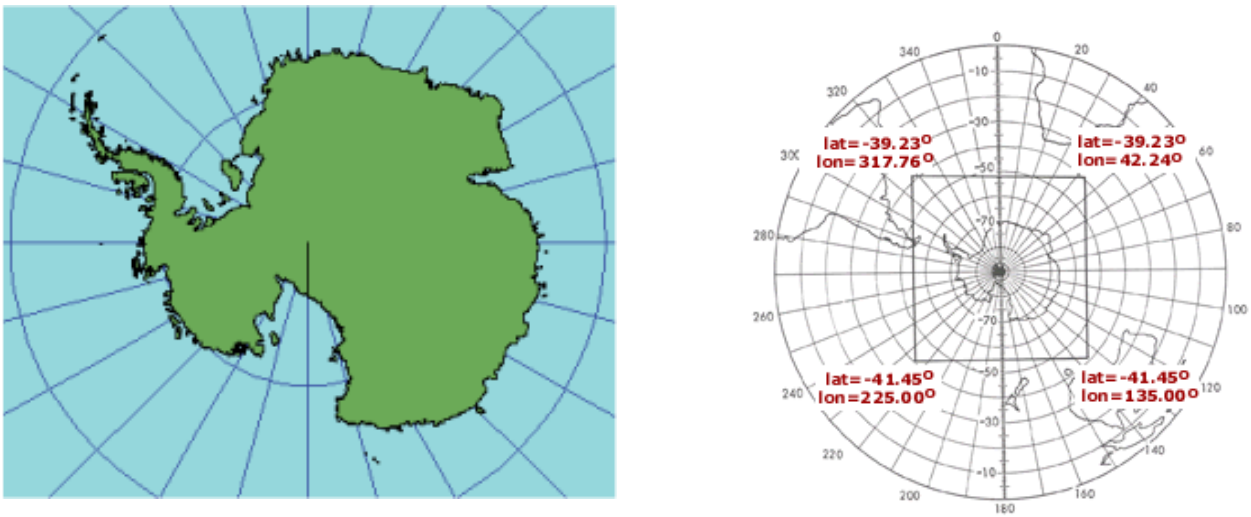

Figure 1: Spatial coverage and projection of the IFREMER SEAICE_ANT_PHY_L3_MY_011_018 product

IV.4 Update Time

SEAICE_ANT_PHY_L3_MY_011_018 products: the fields are automatically updated monthly for those which are made from data using satellites that are still in operation i.e. :

- The products are updated monthly (at the beginning of each month)

cmems_obs-si_ant_physic_my_drift-amsr_P2D
cmems_obs-si_ant_physic_my_drift-amsr_P3D

- The CFOSAT product datasets are covering the 2020-2022 period. They are not updated.

IV.5 Temporal extend of analysis and forecast stored on delivery mechanism

SEAICE_ANT_PHY_L3_MY_011_018 AMSR products were made from April until October. The production will be updated monthly (for month M-2).

IV.6 Other information: mean centre of product, land mask value, missing value

- AMSR datasets :

The 2 day-lag products are an estimation of the mean drift between day 1 and day 3, which means that the mean velocity field is estimated with a center at 12:00 a.m on day 2.

The 3 day-lag products are an estimation of the mean drift between day 1 and day 4, which means that the mean velocity field is estimated with a center at 12:00 a.m on day 3.

- CFOSAT datasets :

The 2 day-lag products are an estimation of the mean drift between day 1 and day 3, which means that the mean velocity field is estimated with a center at 00:00 a.m on day 2.

The 3 day-lag products are an estimation of the mean drift between day 1 and day 4, which means that the mean velocity field is estimated with a center at 00:00 a.m on day 3.

The 30 day-lag products are an estimation of the mean drift between day 1 and day 30, which means that the mean velocity field is estimated with a center at 00:00 a.m on day 16.

V VALIDATION AND ASSESSMENT/ QUALITY CONTROL DONE ON PRODUCTS

The product has been validated over one winter. The description of the validation can be found at https://cersat.ifremer.fr/

Girard-Ardhuin, F., R. Ezraty, 2012 : **Enhanced Arctic sea ice drift estimation merging radiometer and scatterometer data**. *IEEE Trans. Geosci. Remote Sensing,* vol. 50 (7), July 2012, pp 2639-2648. Doi : 10.1109/TGRS.2012.2184124.

VI. ANNEX

Structure of files

```
dimensions:
        y = 720 ;
        x = 60 ;
        time = 1 ;

variables:
        int64 time(time) ;
          time:_FillValue = -9223372036854775808LL ;
          time:axis = "T" ;
```

```
    time:standard_name = "time" ;
    time:units = "days since 1980-01-01 00:00:00" ;
    time:calendar = "proleptic_gregorian" ;
    time:long_name = "reference time of field" ;
    time:authority = "CF-1.7" ;
    time:coverage_content_type = "coordinate" ;
    time:bounds = "time_bnds" ;

  double time_bnds(time, nv) ;
    time_bnds:units = "seconds since 1980-01-01 00:00:00" ;

  double lat(y,x) ;
    lat:long_name = "latitude of the start location of the displacement vector" ;
    lat:standard_name = "latitude" ;
    lat:units = "degrees_north" ;
    lat:_FillValue = 1.e+20 ;
```

```
    lat:authority = "CF-1.7" ;

    lat:valid_range = -39., -89.9. ;

    lat:coverage_content_type = "coordinate" ;

    lat:authority = "CF-1.7" ;


  double lon(y,x) ;

    lon:long_name = "longitude of the start location of the displacement vector" ;

    lon:standard_name = "longitude" ;

    lon:units = "degrees_east" ;

    lon:_FillValue = 1.e+20 ;

    lon:units = "degrees_east" ;

    lon:authority = "CF-1.7" ;

    lon:valid_range = -179.7., 180. ;

    lon:coverage_content_type = "coordinate" ;


  double (eastward_sea_ice_velocity(time, y, x) ;
```

eastward_sea_ice_velocity:long_name = "eastward_sea_ice_velocity" ;

eastward_sea_ice_velocity:units = "m/s" ;

eastward_sea_ice_velocity:valid_min = -0.3858 ;

eastward_sea_ice_velocity:valid_max = 0.3858 ;

eastward_sea_ice_velocity:_FillValue = 1.e+20 ;

eastward_sea_ice_velocity:standard_name = "eastward_sea_ice_velocity" ;

eastward_sea_ice_velocity:authority = "CF-1.7" ;

double (northward_sea_ice_velocity(time, y, x) ;

northward_sea_ice_velocity:long_name = "northward sea ice velocity" ;

northward_sea_ice_velocity:units = "m/s" ;

northward_sea_ice_velocity:valid_min = -0.3858 ;

northward_sea_ice_velocity:valid_max = 0.3858 ;

northward_sea_ice_velocity:_FillValue = 1.e+20 ;

northward_sea_ice_velocity:standard_name = "eastward_sea_ice_velocity" ;

northward_sea_ice_velocity:authority = "CF-1.7" ;

global attributes:

:Conventions = "CF-1.7, ACDD-1.3, ISO 8601" ;

:Metadata_Conventions = "Climate and Forecast (CF) 1.7, Attribute Convention for Data Discovery (ACDD) 1.3" ;

:_NCProperties = "version=2,netcdf=4.7.4,hdf5=1.12.0," ;

:standard_name_vocabulary = "Climate and Forecast (CF) Standard Name Table v79" ;

:naming_authority = "fr.ifremer.cersat" ;

:netcdf_version_id = "4.7.4 of Oct 31 2021 03:14:43 $" ;

:title = "Arctic Ocean - Sea Ice – 2 days Drift, AMSR" ;

:id = "CERSAT-L3-ANT-PSIDRIFT-AMSR-2D-031" ;

:cmems_product_id = "SEAICE_ANT_SEAICE_L3_REP_OBSERVATIONS_011_018" ;

:summary = "This dataset provide a times series of daily sea ice drift over antarctic from AMSR sensor, every 24 hours. It is produced for the Copernicus Marine Service." ;

:references = "Sea Ice drift in the central Arctic using the 89 GHz brightness temperature of the advanced microwave scanning radiometer. Robert Ezraty, Fanny Girard-Ardhuin and Denis Croizé-Fillon, 2007" ;

:metadata_link = "None" ;

:doi = "None" ;

:processing_level = "L3" ;

:time_coverage_resolution = "P1D" ;

:time_coverage_duration = "P3D" ;

:temporal_resolution = "1 day" ;

:key_variables = "northward_sea_ice_velocity, eastward_sea_ice_velocity" ;

:keywords = "Oceans > Sea Ice > Sea ice dynamics" ;

:keywords_vocabulary = "NASA Global Change Master Directory (GCMD) Science Keywords" ;

:project = "Copernicus Marine Service" ;

:program = "Copernicus" ;

:institution = "Institut Francais de Recherche pour l\'Exploitation de la mer / Centre d\'Exploitation et de Recherche Satellitaire" ;

:institution_abbreviation = "Ifremer/CERSAT" ;

:license = "These data are available free of charge under the CMEMS data policy, refer to http://marine.copernicus.eu/services-portfolio/service-commitments-and-licence/" ;

:acknowledgment = "The development of this dataset was funded by Copernicus Marine Service" ;

:citation = ""Ifremer / CERSAT. 2022. Antarctic Sea-ice Drift velocity Vectors from AMSR2 Gridded Level 4 Daily dataset. Ver. 1.0. Ifremer, Plouzane, France. Dataset accessed [YYYY-MM-DD]." ;" ;

:contact = "fanny.ardhuin@ifremer.fr;jfpiolle@ifremer.fr" ;

:technical_support_contact = "cersat@ifremer.fr" ;

:scientific_support_contact = "fanny.ardhuin@ifremer.fr" ;

:creator_email = "fanny.ardhuin@ifremer.fr" ;

:creator_type = "person" ;

:creator_institution = "Ifremer / CERSAT" ;

:creator_name = "Fanny Ardhuin" ;

:creator_url = "http://cersat.ifremer.fr" ;

:publisher_institution = "Ifremer / CERSAT" ;

:publisher_name = "Ifremer / CERSAT" ;

:publisher_url = "https://cersat.ifremer.fr" ;

:processing_software = "" ;

:source = "" ;

:source_version = "" ;

:format_version = "2.0" ;

:product_version = "1.0" ;

:doi = copernicus marine doi ;

:key_variables = "eastward_sea_ice_velocity,northward_sea_ice_velocity" ;

:platform = "GCOM" ;

:platform_type = "Sun-synchronous" ;

:platform_vocabulary = "CEOS mission table" ;

:instrument = "amsr2" ;

:instrument_type = "Imaging multi-spectral MW radiometers" ;

:instrument_vocabulary = "CEOS instrument table" ;

:polarization = "horizontal, vertical" ;

:geospatial_bounds = "POLYGON ((-180.0 -80.0, 180.0 -80.0, 180.0 80.0, -180.0 80.0, -180.0 -80.0))" ;

:geospatial_bounds_crs = "EPSG:4326" ;

:geospatial_bounds_vertical_crs = "EPSG:5831" ;

:geospatial_lat_max = 80. ;

:geospatial_lat_min = -80. ;

:geospatial_lat_resolution = 0.1 ;

:geospatial_lat_units = "degrees_north" ;

:geospatial_lon_max = 180. ;

:geospatial_lon_min = -180. ;

:geospatial_lon_resolution = 0.1 ;

:geospatial_lon_units = "degrees_east" ;

:geospatial_vertical_min = 0. ;

:geospatial_vertical_max = 0. ;

:time_coverage_start = "2022-04-20T12:00:00" ;

:time_coverage_end = "2022-04-21T12:00:00" ;

:spatial_resolution = "0.1 degree" ;

:geospatial_vertical_units = "meters above mean sea level" ;

:date_created = "2023-06-15T08:30:41Z" ;

:date_modified = "2023-06-15T08:30:41Z" ;

:cdm_data_type = "grid" ;

}